UNIVERSITY EXAMINATIONS

UNIVERSITEITSEKSAMENS

UNISA university of south africa

MAT1503

October/November 2018

Linear Algebra

Duration 2 Hours

100 Marks

EXAMINERS
FIRST DR M KOROSTENSKI-DAVIES
SECOND DR L GODLOZA

Closed book examination

This paper consists of 4 pages

THE USE OF A POCKET CALCULATOR IS NOT PERMITTED.

Answer all the questions

There is a total of 100 marks

QUESTION 1

(a) Consider the following system of linear equations

$$\begin{aligned} x - 4y + 6z &= 3 \\ 2x \qquad + 3z &= 1 \\ -2x + 8y - 12z &= -6 \end{aligned}$$

Reduce the augmented matrix for the system to row-echelon form, and then find the solution set of the system. (8)

(b) Use elementary row operations, find a condition on λ so that the system

$$\begin{aligned} 3x + 2y + z &= 0 \\ 6x + 4y + \lambda z &= 0 \end{aligned}$$

is consistent (4)

(c) Show that if A, B and $A + B$ are invertible matrices then

$$A\left(A^{-1} + B^{-1}\right)\, B\,(A + B)^{-1} = I \qquad (5)$$

(d) Use Cramer's Rule to solve the system

$$\begin{bmatrix} -2 & 3 & -1 \\ 1 & 2 & -1 \\ -2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} \qquad (8)$$

[25]

QUESTION 2

(a) Matrices A and B are called **similar** if there exists an invertible matrix P such that

$$A = PBP^{-1}$$

Show that $\det A = \det B$ (4)

(b) Let

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix} \text{ and } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

(i) Find $\det A$ by expanding along the first column and conclude that A is invertible. (4)

(ii) Determine A^{-1} (6)

(iii) Solve the equation $AX = B$ using part (ii). (5)

(iv) Using the result in part (i), write down the value of $\det B$, $\det C$ and $\det D$ where

$$B = \begin{bmatrix} 2 & -4 & 4 \\ 4 & 2 & 2 \\ 2 & 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -2 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$ (6)

[25]

QUESTION 3

Let $\underline{u} = \langle 1, 0, -1 \rangle$ and $\underline{v} = \langle 0, -1, 1 \rangle$ be two vectors.
Determine

(a) the distance between $\underline{u}$ and $\underline{v}$ (3)

(b) the area of the parallelogram bounded by $\underline{u}$ and $\underline{v}$ (5)

(c) (i) $\text{Proj}_{\underline{v}}\underline{u}$, the orthogonal projection of $\underline{u}$ on $\underline{v}$

(ii) the component of $\underline{u}$ orthogonal to $\underline{v}$ (6)

(d) the cartesian equation of the plane containing $\underline{u}$ and $\underline{v}$ (6)

(e) the cosine of the angle θ between $\underline{u}$ and $\underline{v}$ (5)

[25]

QUESTION 4

(a) Consider the points $A(-1, 0, 2)$, $B(2, 0, -1)$ and $C(0, -2, 1)$

Determine

(i) the parametric equation of the line passing through A and B. (4)

(ii) the equatioin of the plane passing through C and perpendicular to the line passing through A and B (3)

(b) Use de Moivre's Theorem to find $(1 + i)^5$ in the form $x + iy$. (5)

(c) Determine all complex cube roots of $-\dfrac{1}{i}$ in polar form. (9)

[TURN OVER]

(d) Let $z_1 = 1 + \sqrt{3}i$ and $z_2 = \sqrt{3} + i$ Find $\dfrac{z_1}{z_2}$ and write your answer in standard form. (4)

[25]

TOTAL [100]